Participant-Independent Identification of Self-Reported Stress Events From Wearable Sensor Data During Alcohol Use Disorder Recovery

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Abstract

Background: Wearable sensors may capture short-term physiological changes related to stress during alcohol use disorder recovery.

Objective: Test whether changes from a person's recent physiological baseline distinguish periods before self-reported stress markers from matched control periods in unseen participants.

Methods: This secondary analysis used the public ADARP dataset and leave-one-participant-out validation.

Results: The 5-minute random forest achieved a participant-mean receiver operating characteristic area under the curve (ROC AUC) of 0.774 (95% CI 0.694-0.851).

Conclusion: Short-horizon wearable changes supported proof of concept for stress-state sensing, not craving, alcohol-use, or relapse prediction.

Introduction

Stress can increase craving and relapse vulnerability during recovery. Wrist-worn sensors measure movement, heart rate, electrodermal activity, and skin temperature throughout daily life. These signals may reveal physiological changes that a periodic survey misses.

Wearable data also vary because of physical activity, environmental temperature, sensor contact, and individual physiology. A model can appear accurate if data from the same person occur in both training and testing. This study therefore kept each test participant completely separate from the training data.

Research question
Can baseline-relative wearable features distinguish periods immediately before self-reported stress markers from matched controls in participants excluded from model training?

Study aims
1. Compare 5-, 15-, 30-, and 60-minute monitoring windows.
2. Compare wearable signal sets.
3. Identify the features that contribute most to held-out performance.

The analysis evaluates retrospective event-window discrimination. It does not test prospective warnings or clinical deployment.

Primary hypothesis
The shortest monitoring window would best capture rapid physiological changes near a stress marker, while longer windows would dilute those changes with unrelated activity.

Materials

Dataset
Public Alcohol and Drug Abuse Research Program (ADARP) dataset.

Participants
11 adults receiving treatment for moderate-to-severe alcohol use disorder.

Wearable signals
Heart rate, electrodermal activity, skin temperature, and triaxial acceleration from an Empatica E4 wrist sensor.

Analytic sample
207 stress-related periods matched to 207 control periods for each monitoring window, producing 414 observations per window.

Predictors
24 baseline-relative features representing each signal's mean, variability, median, minimum, maximum, and trend.

Methodology

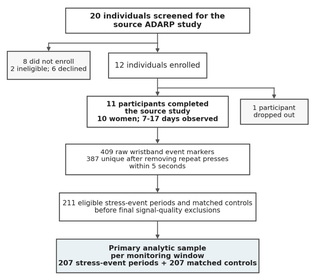


Figure 1. Participant and analytic-sample flow.

Event definition
Each participant-generated E4 stress marker defined a positive observation. A participant-matched unmarked period served as the control.

Feature construction
For each 5-, 15-, 30-, and 60-minute pre-marker window, features measured change from the participant's recent baseline.

Models
Logistic regression and random forest classifiers were compared.

Validation
Eleven-fold leave-one-participant-out validation held out one person at a time. Preprocessing used only the training fold.

Evaluation
The primary metric was ROC AUC. Confidence intervals came from participant-level bootstrap resampling. Modality ablation and held-out permutation importance assessed signal contribution.

Results

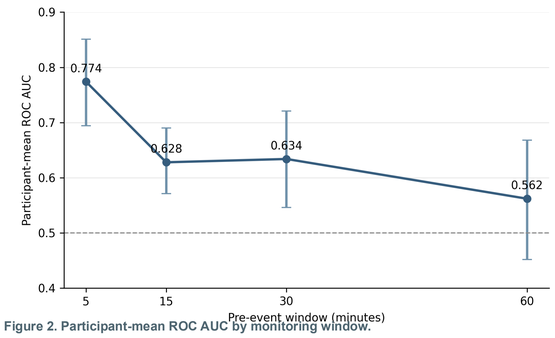


Figure 2. Participant-mean ROC AUC by monitoring window.

Best model: 5-minute random forest
Participant-mean ROC AUC 0.774 (95% CI 0.694-0.851)
Pooled AUC 0.759 | Average precision 0.772 | Accuracy 0.686 | F1 0.688

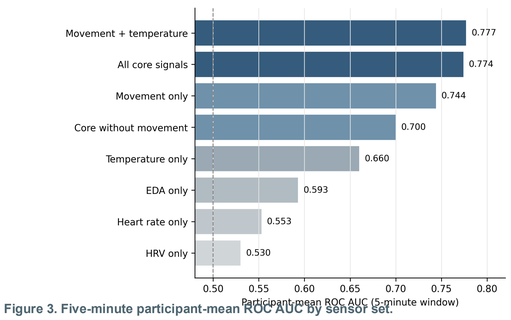


Figure 3. Five-minute participant-mean ROC AUC by sensor set.

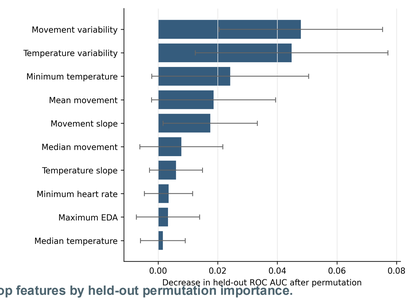


Figure 4. Top features by held-out permutation importance.

Movement plus temperature matched the full signal set (AUC 0.777 versus 0.774). Movement alone reached 0.744, while excluding movement reduced AUC to 0.700. Movement variability and temperature variability ranked highest. Minimum skin temperature was lower before stress markers in 10 of 11 participants. These patterns indicate model reliance, not causality.

Conclusion

Baseline-relative wearable features provided moderate participant-independent discrimination immediately before self-reported stress markers. The 5-minute model outperformed longer windows, suggesting that detectable changes concentrated near the marker rather than across the preceding hour.

Movement and temperature contributed most, although physical activity and environmental conditions may partly explain these patterns.

This study supports proof of concept for short-term wearable stress-state sensing. It does not establish prospective stress onset, craving, alcohol-use, relapse prediction, or readiness for clinical use.

Recommendations

Future studies should:

1. Recruit a larger and more diverse cohort and validate the model in an independent dataset.

2. Collect precisely timed craving, perceived-risk, verified substance-use, intervention, and contextual labels.

3. Evaluate prospective lead time, calibration, alert precision, and false alerts per day.

4. Separate stress-related physiology from exercise, ambient temperature, sleep, medication, and sensor-contact effects.

5. Test personalized baselines and multimodal models while preserving strict participant-level separation.

Clinical decisions should not rely on this proof-of-concept model.

Acknowledgements

BASS Emerging Scholars Program

Data: Sah RK et al. Alcohol and Drug Abuse Research Program wearable dataset. Zenodo. doi:10.5281/zenodo.6640290.

Selected references
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